

YEAR 7 MATHEMATICS · CHAPTER 5

Algebra and equations

5.01 The commutative and associative laws

5.03 Variables and algebraic notation

5.05 Substitution

5.08 Multiplying terms

5.10 Guess, check and improve

5.12 Solving two-step equations

5.02 The distributive law

5.04 Expressions from words

5.07 Adding and subtracting terms

5.09 Dividing terms

5.11 Solving one-step equations

5.13 Equations and problem solving

Name

Class

Teacher

How to use this booklet

- Each lesson starts with **learning intentions** and **success criteria**. Write the **key terms** as your teacher explains them, and use the **notes** box.
- Work through the levels in order: Level 1 , then Level 2, then Level 3. Extension questions stretch you further.
- Some questions ask you to **draw** a diagram, graph or construction. Use the space given, a ruler and a sharp pencil.
- At the end of each lesson, **show off your skill**: choose one question, then tear off the ticket and hand it in.
- Missed a lesson? Each lesson ends with a **summary** that has a worked example for every level. Start there, then try the questions.

WE DO

Grey boxes with a dark border. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box and write your answer.

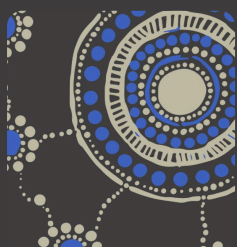
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In this chapter you will

- use the commutative, associative and distributive laws
- use variables and algebraic notation
- write expressions from words and substitute into them
- simplify by adding, subtracting, multiplying and dividing terms
- solve one-step and two-step equations
- solve problems using equations

NSW Mathematics K-10 Syllabus (2022), Stage 4 Number and algebra: MA4-ALG-C-01 generalises number properties to operate with algebraic expressions, and MA4-EQU-C-01 solves linear equations.



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
5.01

The commutative and associative laws

Date _____

Learning intentions

We are learning to:

- use the commutative and associative laws to make calculations easier

Success criteria

I can:

- ☐ know that order does not matter for + and $\times$ (commutative law)
- ☐ know that grouping does not matter for + and $\times$ (associative law)
- ☐ regroup numbers to make friendly pairs such as 10s and 100s

Key terms Write the definition as your teacher explains it.**Commutative****Associative**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** $7 + 18 + 3$ **Example 2** $91 + 18 + 9$ **YOU DO** On your own: Set A. Evaluate by pairing friendly numbers.1 $7 + 18 + 3$

2 $91 + 18 + 9$

3 $46 + 25 + 4$

4 $15 + 38 + 5$

5 $64 + 17 + 36$

6 $8 + 29 + 12 + 1$

YOU DO On your own: Set B. Evaluate by pairing friendly numbers.

1 $2 \times 17 \times 5$

4 $4 \times 13 \times 25$

2 $25 \times 31 \times 4$

5 $50 \times 7 \times 2$

3 $5 \times 9 \times 20$

6 $2 \times 8 \times 5 \times 5$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** $2 \times 17 \times 5$ **Example 4** $25 \times 31 \times 4$ **YOU DO** On your own: Set C. Evaluate mentally.

1 13×30

3 12×50

5 600×4

2 7×400

4 30×80

6 25×12



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** 13×30

Example 6 7×400

YOU DO On your own: Set D. Think it through.

- 1 Does the commutative law work for subtraction? Test it with $10 - 4$.

- 3 Evaluate $125 \times 17 \times 8$ mentally.

- 2 Does the associative law work for division? Test it with $(24 \div 6) \div 2$.

- 4 Evaluate $1 + 2 + 3 + \dots + 20$ by pairing numbers.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Evaluate $16 + 27 + 4$.
Level 2Evaluate $5 \times 23 \times 2$.
Level 3Evaluate $20 \times 13 \times 5$.

Extension · Stretch yourself

Worked example

Evaluate $4 \times 7 \times 25 \times 3 \times 5 \times 2$ mentally.

1. Regroup: $(4 \times 25) \times (5 \times 2) \times 7 \times 3$
2. $100 \times 10 \times 21$

Answer: 21 000

YOU DO

On your own: read the worked example, then try these.

- E1** Evaluate $8 \times 125 \times 9 \times 5 \times 2$ mentally.

- E2** Explain why $38 + 47 + 62 + 53$ is easy to do mentally.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Commutative: $a + b = b + a$ and $a \times b = b \times a$.
 2. Associative: $(a + b) + c = a + (b + c)$.
 3. Look for pairs making 10, 100 or 1000.
 4. Not true for $-$ and $\div$.

Level 1

$$7 + 18 + 3$$

1. $(7 + 3) + 18$

Answer: 28

Level 2

$$25 \times 31 \times 4$$

1. $(25 \times 4) \times 31$

Answer: 3100

Level 3

$$13 \times 30$$

1. $13 \times 3 \times 10$

Answer: 390

Extension

$$4 \times 7 \times 25 \times 3 \times 5 \times 2$$

1. $100 \times 10 \times 21$

Answer: 21 000

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LESSON
5.02

The distributive law

Date _____

Learning intentions

We are learning to:

- use the distributive law for mental arithmetic

Success criteria

I can:

- ☐ split a number into tens and units, or a friendly number $\pm$ a bit
- ☐ multiply each part, then add or subtract
- ☐ choose the easiest way to split

Key terms Write the definition as your teacher explains it.**Distributive law****Expand**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** 22×9 as $22 \times (10 - 1)$ **Example 2** 17×8 as $(10 + 7) \times 8$ **YOU DO** On your own: Set A. Complete the working to evaluate.

1 $14 \times 6 = 10 \times 6 + 4 \times 6$

2 $23 \times 4 = 20 \times 4 + 3 \times 4$

3 $31 \times 7 = 30 \times 7 + 1 \times 7$

4 $19 \times 5 = 20 \times 5 - 1 \times 5$

5 $42 \times 3 = 40 \times 3 + 2 \times 3$

6 $29 \times 4 = 30 \times 4 - 1 \times 4$

YOU DO On your own: Set B. Use the distributive law to evaluate.

1 22×9

4 13×12

2 17×8

5 15×7

3 27×11

6 39×3

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** 27×11 **Example 4** 13×12 **YOU DO** On your own: Set C. Use the distributive law. Show your working.

1 48×6

3 101×23

5 8×199

2 99×35

4 53×12

6 65×21



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO

With your teacher: copy what they write on the board.

Example 5

48×6

Example 6

99×35

YOU DO

On your own: Set D. Think it through.

- 1 Evaluate $7 \times 68 + 7 \times 32$ mentally.

- 3 Show two different ways to evaluate 18×25 .

- 2 Evaluate $15 \times 43 - 15 \times 33$ mentally.

- 4 Explain why $6 \times 97 = 6 \times 100 - 6 \times 3$.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO**Show off your skill**Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1Evaluate 12×7 using the distributive law.**Level 2**Evaluate 19×6 .**Level 3**Evaluate 102×45 .

Extension · Stretch yourself

Worked example

Evaluate 999×47 mentally.

- $999 = 1000 - 1$
- $47\ 000 - 47$

Answer: 46 953

YOU DO

On your own: read the worked example, then try these.

- E1** Evaluate 1001×64 mentally.

- E2** Evaluate $37 \times 99 + 37$ mentally.

Lesson summary

Missed the lesson? Start here.

- Remember**
- $a \times (b + c) = a \times b + a \times c$.
 - Or near a friendly number: $9 = 10 - 1$.

- Split into tens and units: $17 = 10 + 7$.
- Multiply each part, then add or subtract.

Level 1

$$22 \times 9$$

- $22 \times 10 - 22$

Answer: 198

Level 2

$$17 \times 8$$

- $80 + 56$

Answer: 136

Level 3

$$99 \times 35$$

- $3500 - 35$

Answer: 3465

Extension

$$999 \times 47$$

- $47\ 000 - 47$

Answer: 46 953

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LESSON
5.03

Variables and algebraic notation

Date _____

Learning intentions

We are learning to:

- use a variable to describe a pattern
- write and read algebraic notation

Success criteria

I can:

- ☐ write a general rule using a variable
- ☐ simplify notation, e.g. $5 \times a \times 2 = 10a$
- ☐ write a term in expanded form

Key terms Write the definition as your teacher explains it.

Variable

Pronumeral

Term

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

Write the rule for: $3 - (-3) = 2 \times 3$, $5 - (-5) = 2 \times 5$, $11 - (-11) = 2 \times 11$

Example 2

Simplify $x + x + x + x$.

YOU DO

On your own: Set A. Simplify.

1 $x + x + x + x$

2 $u \div 5$

3 $2y - y$

4 $5 \times a \times 2$

5 $m \times 7$

6 $p \times q$

YOU DO On your own: Set B. Simplify.

① $8 \times c \times b \times 3 \times a$

④ $3 \times y \times y$

② $2 \times m \times w \times 10 \times w$

⑤ $a + a + b + b + b$

③ $k \times k \times k$

⑥ $4 \times d \div 2$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Simplify $5 \times a \times 2$.**Example 4** Simplify $8 \times c \times b \times 3 \times a$.**YOU DO** On your own: Set C. Write in expanded form.

① $4mnk$

③ $9c^2$

⑤ a^3b

② $\frac{6a}{b}$

④ $3xy^2$

⑥ $\frac{5m}{2n}$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Write $4mnk$ in expanded form.
Example 6 Write $9c^2$ and $\frac{6a}{b}$ in expanded form.
YOU DO On your own: Set D. Patterns and rules.

- 1 Write a rule for: $4 + 4 = 2 \times 4$, $7 + 7 = 2 \times 7$, $10 + 10 = 2 \times 10$

- 3 Matchstick squares: 1 square uses 4 sticks, 2 squares use 7, 3 squares use 10. Find a rule for n squares.

- 2 Write a rule for: $3 \times 0 = 0$, $8 \times 0 = 0$, $15 \times 0 = 0$

- 4 Is $3a^2$ the same as $(3a)^2$? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1Simplify $y + y + y$.
Level 2Simplify $4 \times p \times 3 \times q$.
Level 3Write $5a^2b$ in expanded form.

Extension · Stretch yourself

Worked example

Write a rule for the pattern: $1 \times 3 = 2^2 - 1$, $2 \times 4 = 3^2 - 1$, $3 \times 5 = 4^2 - 1$

1. First number is n , second is $n + 2$
2. The square is of $n + 1$

Answer: $n(n + 2) = (n + 1)^2 - 1$

YOU DO

On your own: read the worked example, then try these.

- E1** Write a rule for: $5 \times 5 - 4 \times 6 = 1$, $8 \times 8 - 7 \times 9 = 1$, $3 \times 3 - 2 \times 4 = 1$

- E2** Test your rule with a number of your own.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. A **variable** is a letter that stands for a number.
 2. Leave out $\times$: $3 \times a = 3a$. Number first, letters in order.
 3. $a \times a = a^2$; $a \div 5 = a/5$.
 4. Expanded form shows every $\times$.

Level 1

Rule for $3 - (-3) = 2 \times 3 \dots$

1. Replace the number with a

Answer: $a - (-a) = 2a$

Level 2

$5 \times a \times 2$

1. $5 \times 2 = 10$

Answer: $10a$

Level 3

$2 \times m \times w \times 10 \times w$

1. $20 \times m \times w^2$

Answer: $20mw^2$

Extension

$9c^2$

1. $9 \times c \times c$

Answer: $9 \times c \times c$

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LESSON
5.04

Expressions from words

Date _____

Learning intentions

We are learning to:

- write algebraic expressions from worded statements

Success criteria

I can:

- ☐ know the words for +, −, × and ÷ (sum, difference, product, quotient)
- ☐ write an expression using a variable
- ☐ use brackets when needed

Key terms Write the definition as your teacher explains it.**Expression****Consecutive**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** 3 times the number N **Example 2** the difference between N and 5**YOU DO** On your own: Set A. Write an expression. Use N for the number.

1 the number plus 7

2 3 times the number

3 the number minus 5

4 the number divided by 4

5 the next consecutive number

6 one-third of the number

YOU DO On your own: Set B. Write an expression.1 the sum of M and 34 the product of a and b 2 5 more than B 5 10 less than y 3 $2H$ decreased by k 6 p shared equally among 6**Level 2 · Building up****WE DO** With your teacher: copy what they write on the board.**Example 3** the next consecutive number after N **Example 4** one-third of N **YOU DO** On your own: Set C. Write an expression.1 double the sum of x and 43 7 less than 3 times w 5 the number before n 2 the square of t 4 half of the difference between m and 16 the average of a and b



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $2H$ decreased by k **Example 6** double the sum of x and 4
YOU DO On your own: Set D. Write an expression for each situation.

- 1 The cost of n pens at \$3 each.

- 3 A rectangle is L cm long and 5 cm wide. Write its perimeter.

- 2 Sam is s years old. His sister is 4 years older. Find the sum of their ages.

- 4 The change from \$50 after buying k drinks at \$2.50 each.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Write “the number plus 12” using N .
Level 2Write “4 less than y ”.
Level 3Write “the product of m and 5, then add 2”.

Extension · Stretch yourself

Worked example

Write an expression for the sum of three consecutive numbers starting at n , and simplify it.

- $n + (n + 1) + (n + 2)$
- Collect like terms

Answer: $3n + 3$

YOU DO

On your own: read the worked example, then try these.

- E1** Write and simplify the sum of three consecutive even numbers starting at e .

- E2** Explain why the sum of 3 consecutive numbers is always divisible by 3.

Lesson summary

Missed the lesson? Start here.

- Remember**
- Sum $\rightarrow +$, difference $\rightarrow -$.
 - Product $\rightarrow \times$, quotient $\rightarrow \div$.
 - More than $\rightarrow +$, less than $\rightarrow -$ (order!).
 - Use brackets when a whole group is multiplied.

Level 1

3 times N

- $3 \times N$

Answer: $3N$

Level 2

N decreased by 5

- $N - 5$

Answer: $N - 5$

Level 3

Double the sum of x and 4

- Brackets first

Answer: $2(x + 4)$

Extension

3 consecutive numbers

- $n + n + 1 + n + 2$

Answer: $3n + 3$

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LESSON
5.05

Substitution

Date _____

Learning intentions

We are learning to:

- find the value of an expression by substituting numbers

Success criteria

I can:

- ☐ replace each variable with its value, using brackets
- ☐ follow the order of operations
- ☐ substitute into a formula

Key terms Write the definition as your teacher explains it.

Substitute

Evaluate

Formula

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** If $a = 2$ and $b = 5$, find $a + b$.**Example 2** If $a = 2$ and $c = 6$, find $3a - c$.**YOU DO** On your own: Set A. If $a = 2$, $b = 5$ and $c = 6$, evaluate:1 $a + b$

4 bc

2 $c - a$

5 $a + b + c$

3 $4b$

6 $3a - c$

YOU DO On your own: Set B. If $a = 2$, $b = 5$ and $c = 6$, evaluate:

1 $a + 2b - c$

4 $\frac{c}{a}$

2 abc

5 $2(a + b)$

3 c^2

6 $b^2 - c$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** If $a = 2$, $b = 5$, $c = 6$, find abc .**Example 4** If $C = 30h + 19$, find C when $h = 3$.**YOU DO** On your own: Set C. If $C = 30h + 19$, find C when:

1 $h = 3$

3 $h = 5$

5 $h = 0$

2 $h = -1$

4 $h = 2.5$

6 $h = 10$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Find C when $h = -1$.
Example 6 Find C when $h = 2.5$.
YOU DO On your own: Set D. Substitute into each formula.**1** $P = 2L + 2W$. Find P when $L = 8$ and $W = 3$.
3 $F = 1.8C + 32$. Find F when $C = 25$.
2 $A = \frac{1}{2}bh$. Find A when $b = 10$ and $h = 7$.
4 If $x = -3$, evaluate $x^2 + 2x$.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1If $m = 4$, evaluate $5m$.
Level 2If $x = 3$ and $y = 7$, evaluate $2x + y$.
Level 3If $A = l - 3$, find A when $l = -2$.

Extension · Stretch yourself

Worked example

A taxi charges $C = 4 + 2.5k$ dollars for k km. How far can you go for \$39?

1. $39 = 4 + 2.5k$
2. $35 = 2.5k$
3. $k = 35 \div 2.5$

Answer: 14 km

YOU DO

On your own: read the worked example, then try these.

- E1 Using the same formula, find the cost of a 22 km trip.

- E2 A phone plan costs $C = 15 + 0.2m$ for m minutes. How many minutes for \$27?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Write the expression.
 2. Replace each variable with its value in brackets.
 3. Use the order of operations.
 4. Remember $3a$ means $3 \times a$.

Level 1

$$a + b, a = 2, b = 5$$

1. $2 + 5$

Answer: 7

Level 2

$$3a - c$$

1. $3 \times 2 - 6$

Answer: 0

Level 3

$$C = 30h + 19, h = -1$$

1. $30 \times (-1) + 19$

Answer: -11

Extension

$$\text{Taxi for } \$39$$

1. $2.5k = 35$

Answer: 14 km

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LESSON
5.07

Adding and subtracting terms

Date _____

Learning intentions

We are learning to:

- simplify expressions by collecting like terms

Success criteria

I can:

- ☐ identify **like terms** (same letters, same powers)
- ☐ add and subtract the coefficients of like terms
- ☐ collect several sets of like terms

Key terms Write the definition as your teacher explains it.**Like terms****Coefficient**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1 $3m + 4m$ **Example 2** $12m - 4m$

YOU DO

On your own: Set A. Simplify.

1 $3m + 4m$

2 $12m - 4m$

3 $5a + a$

4 $9k - 9k$

5 $2x + 3x + 4x$

6 $7y - 4y - y$

YOU DO On your own: Set B. Are these like terms? Write yes or no.

1 $3a$ and $5a$

4 m^2 and $3m$

2 $2x$ and $2y$

5 6 and $6p$

3 $4ab$ and $7ba$

6 $-2k^2$ and k^2

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 $7y - 4y - y$

Example 4 $6uv + 4 + 12uv$

YOU DO On your own: Set C. Simplify.

1 $6uv + 4 + 12uv$

3 $8x - 3 + 2x + 7$

5 $10 - 3m - 4 + 5m$

2 $5a + 3b + 2a + b$

4 $4pq - pq + 3qp$

6 $2y + 5 - 2y$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $5h^2 + 6h + 8h^2 + 11h$ **Example 6** $13h^3 - 6h - 8h - 11h^3$ **YOU DO** On your own: Set D. Simplify.**1** $5h^2 + 6h + 8h^2 + 11h$ **3** $3ab - 2a + 5ab + 7a - b$ **2** $13h^3 - 6h - 8h - 11h^3$ **4** A triangle has sides $2x$, $3x + 1$ and $x + 4$. Write and simplify its perimeter.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Simplify $6k + 3k$.
Level 2Simplify $4a + 7 - a + 2$.
Level 3Simplify $3x^2 + 5x - x^2 - 8x$.

Extension · Stretch yourself

Worked example

Simplify $4x^2y - 3xy^2 + x^2y + 5xy^2 - xy$.

1. Like terms: $4x^2y + x^2y$
2. $-3xy^2 + 5xy^2$
3. $-xy$ has no partner

Answer: $5x^2y + 2xy^2 - xy$

YOU DO

On your own: read the worked example, then try these.

- E1 Simplify $7a^2 - 3a + 2 - a^2 + 5a - 8$.

- E2 Find the missing term: $5m + ? - 2m = 9m$

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Like terms have exactly the same letters and powers.
 2. Add or subtract only like terms.
 3. Keep the sign in front of each term.
 4. Write the answer with the highest power first.

Level 1

$$3m + 4m$$

1. $3 + 4 = 7$

Answer: $7m$

Level 2

$$7y - 4y - y$$

1. $7 - 4 - 1$

Answer: $2y$

Level 3

$$6uv + 4 + 12uv$$

1. $6 + 12 = 18$

Answer: $18uv + 4$

Extension

$$5h^2 + 6h + 8h^2 + 11h$$

1. h^2 terms, then h terms

Answer: $13h^2 + 17h$

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LESSON
5.08

Multiplying terms

Date _____

Learning intentions

We are learning to:

- multiply algebraic terms

Success criteria

I can:

- ☐ multiply the numbers first, then the letters
- ☐ write repeated letters as powers
- ☐ use the sign rules for negative terms

Key terms Write the definition as your teacher explains it.**Coefficient****Power**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1 $3m \times 4m$ **Example 2** $a \times a \times a \times a$

YOU DO

On your own: Set A. Simplify.

1 $3 \times 4m$

2 $2a \times 5$

3 $a \times b$

4 $a \times a \times a \times a$

5 $x \times 7 \times y$

6 $3m \times 4m$

YOU DO On your own: Set B. Simplify.

1 $6p \times 2q$

4 $5ab \times 2a$

2 $9v \times 4 \times 5v$

5 $4x \times 3x \times x$

3 $k \times k \times 2$

6 $2c \times 2c \times 2c$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 $9v \times 4 \times 5v$

Example 4 $3pq \times 5pq$

YOU DO On your own: Set C. Simplify. Watch the signs.

1 $-7d \times (-3d)$

3 $3x \times (-x)$

5 $-3x^2y \times 2x$

2 $-2a \times 6b$

4 $-4m \times (-m) \times 2$

6 $5 \times (-f) \times (-f)$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $-7d \times (-3d)$ **Example 6** $lmn \times lmn \times lmn$ **YOU DO** On your own: Set D. Simplify.**1** $3pq \times 5pq$ **3** $2a^2 \times 3a^3$ **2** $lmn \times lmn \times lmn$ **4** A rectangle is $3x$ cm long and $2y$ cm wide. Write its area.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Simplify $5 \times 3k$.
Level 2Simplify $4a \times 6ab$.
Level 3Simplify $-2x \times 3x \times (-y)$.

Extension · Stretch yourself

Worked example

Simplify $(2ab)^3$.

- $2ab \times 2ab \times 2ab$
- Numbers: $2 \times 2 \times 2 = 8$
- a^3b^3

Answer: $8a^3b^3$

YOU DO

On your own: read the worked example, then try these.

E1 Simplify $(-3m)^2$.

E2 Find the missing term: $4x \times ? = 20x^2y$.

Lesson summary

Missed the lesson? Start here.

Remember

1. Multiply the numbers (coefficients).
3. Same letter repeated → use a power.

2. Multiply the letters, in alphabetical order.
4. Signs: same → +, different → -.

Level 1

$$3m \times 4m$$

$$1. 3 \times 4 = 12, m \times m = m^2$$

Answer: $12m^2$

Level 2

$$9v \times 4 \times 5v$$

$$1. 9 \times 4 \times 5 = 180$$

Answer: $180v^2$

Level 3

$$-7d \times (-3d)$$

$$1. - \times - = +$$

Answer: $21d^2$

Extension

$$(2ab)^3$$

$$1. 8, a^3, b^3$$

Answer: $8a^3b^3$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
5.09

Dividing terms

Date _____

Learning intentions

We are learning to:

- divide algebraic terms

Success criteria

I can:

- ☐ write a division as a fraction
- ☐ cancel common factors in the numbers and the letters
- ☐ simplify answers that are still fractions

Key terms Write the definition as your teacher explains it.**Quotient****Cancel**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1 $15m \div 3$ **Example 2** $9v \div v$

YOU DO

On your own: Set A. Simplify.

1 $15m \div 3$

2 $9v \div v$

3 $\frac{49y}{7}$

4 $8ab \div 2$

5 $20k \div 5k$

6 $\frac{6pq}{q}$

YOU DO On your own: Set B. Simplify.

1 $\frac{12x^2y}{4xy}$

4 $24t^3 \div 8t$

2 $18m^2 \div 6m$

5 $\frac{3x}{12}$

3 $\frac{10ab}{5b}$

6 $\frac{4a}{6ab}$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 $\frac{49y}{7}$

Example 4 $\frac{12x^2y}{4xy}$

YOU DO On your own: Set C. Simplify.

1 $32de^2 \div 10de$

3 $\frac{-15x^2}{-5x}$

5 $\frac{14pq^2}{21pq}$

2 $\frac{-20mn}{4m}$

4 $\frac{9a^2b^2}{3ab}$

6 $-36y^3 \div 4y^2$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $32de^2 \div 10de$ **Example 6** $\frac{-3 \times 2y \times 2x}{3x \times (-x)}$ **YOU DO** On your own: Set D. Simplify.

1 $\frac{-3 \times 2y \times 2x}{3x \times (-x)}$

3 $\frac{10m2n}{2m \times 5n}$

2 $\frac{4a \times 6b}{8ab}$

4 A rectangle has area $24xy \text{ cm}^2$ and length $6x \text{ cm}$. Find its width.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Simplify $21p \div 7$.
Level 2Simplify $\frac{16ab2}{4b}$.
Level 3Simplify $\frac{-10 \times 3}{15x}$.

Extension · Stretch yourself

Worked example

Simplify $\frac{6a3b \times 4ab2}{12a2b2}$.

- Top: $24a^4b^3$
- $\div 12a^2b^2$

Answer: $2a^2b$

YOU DO

On your own: read the worked example, then try these.

E1 Simplify $\frac{(3m)2}{6m}$.

E2 Simplify $\frac{8 \times 2y \times 3y}{6xy \times 2x}$.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Write the division as a fraction.
 3. Cancel letters that are on top and bottom.

2. Cancel the numbers by their HCF.
4. Signs: same $\rightarrow +$, different $\rightarrow -$.

Level 1

$15m \div 3$

1. $15 \div 3 = 5$

Answer: $5m$

Level 2

$\frac{12 \times 2y}{4xy}$

1. $12 \div 4 = 3$, cancel x and y

Answer: $3x$

Level 3

$32de^2 \div 10de$

1. $32/10 = 16/5$, cancel d and e

Answer: $\frac{16e}{5}$

Extension

$\frac{6a3b \times 4ab2}{12a2b2}$

1. $24 \div 12 = 2$

Answer: $2a^2b$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
5.10

Guess, check and improve

Date _____

Learning intentions

We are learning to:

- solve equations by guess, check and improve

Success criteria

I can:

- ☐ substitute a guess into the equation
- ☐ decide whether the guess is too high or too low
- ☐ improve the guess until it is correct

Key terms Write the definition as your teacher explains it.**Equation****Solution**

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.**Example 1** Is $x = 4$ a solution of $3x + 2 = 14$?**Example 2** Solve $4m - 9 = 7$.**YOU DO** On your own: Set A. Is the value a solution? Write yes or no.

1 $x + 5 = 12$, $x = 7$

4 $2k + 1 = 11$, $k = 5$

2 $3y = 21$, $y = 6$

5 $\frac{p}{2} = 8$, $p = 4$

3 $m - 4 = 9$, $m = 13$

6 $10 - t = 3$, $t = 7$

YOU DO On your own: Set B. Solve by guess, check and improve.

1 $x + 8 = 15$

4 $2a + 1 = 9$

2 $4m = 36$

5 $3n - 2 = 10$

3 $y - 6 = 11$

6 $\frac{w}{5} = 4$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Solve $\frac{k}{7} + 2 = 2$.**Example 4** Solve $\frac{d}{3} - 4 = 2$.**YOU DO** On your own: Set C. Solve by guess, check and improve. Show your guesses in a table.

1 $4m - 9 = 7$

3 $\frac{d}{3} - 4 = 2$

5 $7b - 10 = 39$

2 $\frac{k}{7} + 2 = 2$

4 $5a + 13 = 58$

6 $3(x + 2) = 27$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** Solve $5a + 13 = 58$.
Example 6 Solve $2x + 3 = 4$ (the answer is not a whole number).
YOU DO On your own: Set D. Solve by guess, check and improve.

1 $2x + 3 = 4$

3 $x^2 + x = 30$

2 $n^2 = 144$

4 I think of a number, multiply by 6 and subtract 5. The answer is 43. What is my number?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Is $a = 3$ a solution of $5a - 1 = 14$?
Level 2Solve $3x + 4 = 25$.
Level 3Solve $\frac{y}{4} + 7 = 12$.

Extension · Stretch yourself

Worked example

Solve $x^2 = 20$ to 1 decimal place by guess, check and improve.

- $4^2 = 16$ (low), $5^2 = 25$ (high)
- $4.4^2 = 19.36$ (low), $4.5^2 = 20.25$ (high)
- $4.47^2 \approx 19.98$, so round to 4.5

Answer: $x \approx 4.5$

YOU DO

On your own: read the worked example, then try these.

- E1 Solve $x^3 = 50$ to 1 decimal place.

- E2 Solve $x(x + 1) = 50$ to 1 decimal place.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Guess a value for the variable.
 2. Substitute it into the equation.
 3. Too big or too small? Adjust your guess.
 4. Stop when both sides are equal.

Level 1

Is $x = 4$ a solution of $3x + 2 = 14$?

- $3 \times 4 + 2 = 14$

Answer: yes

Level 2

$$4m - 9 = 7$$

- $m = 3$: 3 (low)
- $m = 4$: 7 ✓

Answer: $m = 4$

Level 3

$$d/3 - 4 = 2$$

- $d = 18$: $6 - 4 = 2$ ✓

Answer: $d = 18$

Extension

$$x^2 = 20$$

- 4.4 low, 4.5 high

Answer: $x \approx 4.5$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
5.11

Solving one-step equations

Date _____

Learning intentions

We are learning to:

- solve one-step equations algebraically

Success criteria

I can:

- ☐ use the inverse operation to undo $+$, $-$, $\times$ or $\div$
- ☐ do the same to both sides of the equation
- ☐ check the solution by substituting

Key terms Write the definition as your teacher explains it.**Inverse operation****Solve**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

$w + 17 = 52$

Example 2

$e - 12 = 0$

YOU DO

On your own: Set A. Solve.

1 $w + 17 = 52$

2 $e - 12 = 0$

3 $a + 9 = 20$

4 $m - 6 = 14$

5 $x + 25 = 40$

6 $y - 13 = 7$

YOU DO On your own: Set B. Solve.

1 $5p = 15$

4 $\frac{n}{5} = 9$

2 $\frac{x}{2} = 6$

5 $12c = 60$

3 $8t = 72$

6 $\frac{h}{10} = 3$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 $5p = 15$

Example 4 $\frac{x}{2} = 6$

YOU DO On your own: Set C. Solve. Show each step and check.

1 $-4y = 28$

3 $\frac{m}{-3} = 5$

5 $6b = 21$

2 $k + 3.5 = 1$

4 $x - 8 = -12$

6 $0.5q = 7$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** $-4y = 28$ **Example 6** $k + 3.5 = 1$ **YOU DO** On your own: Set D. Write an equation, then solve it.**1** A number plus 19 is 45.**3** A number divided by 4 is 11.**2** Seven times a number is 84.**4** Take 23 from a number and the answer is 58.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Solve $n + 8 = 30$.**Level 2**Solve $9x = 63$.**Level 3**Solve $\frac{t}{6} = -4$.

Extension · Stretch yourself

Worked example

Solve $15 = x - 4$ and check.

1. The variable can be on either side
2. Add 4 to both sides: $19 = x$

Answer: $x = 19$. Check: $19 - 4 = 15$ ✓

YOU DO

On your own: read the worked example, then try these.

E1 Solve $36 = 9n$.

E2 Solve $\frac{2}{3}a = 10$.

Lesson summary

Missed the lesson? Start here.

Remember 1. Ask: what is being done to the variable?
3. $+$ $\leftrightarrow$ $-$, $\times$ $\leftrightarrow$ $\div$.

2. Do the **inverse** to both sides.
4. Check by substituting back.

Level 1

$w + 17 = 52$

1. $- 17$ from both sides

Answer: $w = 35$

Level 2

$5p = 15$

1. $\div 5$

Answer: $p = 3$

Level 3

$x/2 = 6$

1. $\times 2$

Answer: $x = 12$

Extension

$15 = x - 4$

1. $+ 4$

Answer: $x = 19$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
5.12

Solving two-step equations

Date _____

Learning intentions

We are learning to:

- solve two-step equations algebraically

Success criteria

I can:

- ☐ undo + or – first, then $\times$ or $\div$
- ☐ set out each step on a new line
- ☐ solve equations with fractions

Key terms Write the definition as your teacher explains it.**Two-step equation****Check**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

$3t + 4 = 10$

Example 2

$4y - 7 = 5$

YOU DO

On your own: Set A. Solve.

1 $2n + 6 = 12$

2 $3t + 4 = 10$

3 $5a - 2 = 18$

4 $4y - 7 = 5$

5 $6k + 1 = 31$

6 $10b - 5 = 45$

YOU DO On your own: Set B. Solve.

1 $\frac{m}{4} - 8 = 9$

4 $\frac{p}{5} - 3 = 1$

2 $\frac{x}{3} + 5 = 8$

5 $\frac{y}{6} + 2 = 4$

3 $\frac{a}{2} + 1 = 7$

6 $\frac{w}{10} - 1 = 0$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 $\frac{m}{4} - 8 = 9$

Example 4 $\frac{x}{3} + 5 = 8$

YOU DO On your own: Set C. Solve. Show each step.

1 $\frac{m-4}{2} = 3$

3 $7 - 2x = 15$

5 $5d - 8 = -23$

2 $\frac{k+5}{3} = 4$

4 $3x + 11 = 2$

6 $2(x+3) = 18$



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.

Example 5 $\frac{m-4}{2} = 3$

Example 6 $7 - 2x = 15$

YOU DO On your own: Set D. Solve. Show each step.

① $\frac{2x}{3} + 1 = 9$

③ $1.5t + 2 = 8$

② $4(y - 2) = 30$

④ $\frac{3n-1}{4} = 5$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Solve $2x + 5 = 17$.

Level 2

Solve $\frac{n}{3} - 2 = 4$.

Level 3

Solve $3(k - 1) = 24$.

Extension · Stretch yourself

Worked example

Solve $5x + 3 = 2x + 18$.

1. $- 2x$ from both sides: $3x + 3 = 18$
2. $- 3$: $3x = 15$
3. $\div 3$

Answer: $x = 5$

YOU DO

On your own: read the worked example, then try these.

E1 Solve $7a - 4 = 3a + 12$.

E2 Solve $2(x + 5) = x + 14$.

Lesson summary

Missed the lesson? Start here.

Remember

1. Undo $+$ or $-$ first (both sides).
2. Then undo $\times$ or $\div$.
3. For a fraction like $(m - 4)/2$: $\times$ first.
4. Check in the original equation.

Level 1

$3t + 4 = 10$

1. $- 4$: $3t = 6$
2. $\div 3$

Answer: $t = 2$

Level 2

$m/4 - 8 = 9$

1. $+ 8$: $m/4 = 17$
2. $\times 4$

Answer: $m = 68$

Level 3

$(m - 4)/2 = 3$

1. $\times 2$: $m - 4 = 6$
2. $+ 4$

Answer: $m = 10$

Extension

$5x + 3 = 2x + 18$

1. $3x = 15$

Answer: $x = 5$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
5.13

Equations and problem solving

Date _____

Learning intentions

We are learning to:

- write and solve equations to solve problems

Success criteria

I can:

- ☐ choose a variable for the unknown
- ☐ write an equation from the words
- ☐ solve it and answer the question in a sentence

Key terms Write the definition as your teacher explains it.**Unknown****Profit**

Notes

Level 1 · Getting started

WE DO

With your teacher: copy what they write on the board.

Example 1

A number multiplied by 17 is 1003.
Which equation: $N + 17 = 1003$, $1003 - N = 17$, $17N = 1003$ or $1003 + N = 17$?

Example 2

Solve $17N = 1003$.

YOU DO

On your own: Set A. Write an equation for each. Do not solve.

- 1 A number plus 8 is 20.

- 2 Five times a number is 45.

- 3 A number minus 7 is 3.

- 4 Half a number is 16.

- 5 Double a number plus 1 is 15.

- 6 A number times 3, minus 4, is 11.

YOU DO On your own: Set B. Solve each equation.

1 $n + 8 = 20$

4 $\frac{n}{2} = 16$

2 $5n = 45$

5 $2n + 1 = 15$

3 $n - 7 = 3$

6 $3n - 4 = 11$

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.

Example 3 Raffle prizes cost \$180 and tickets are \$5. Write and solve an equation for the number of tickets to cover costs.

Example 4 How many tickets must be sold to make a \$2100 profit?

YOU DO On your own: Set C. Write an equation, solve it, and answer in a sentence.

1 A pen and a \$4 notebook cost \$6.50. Find the price of the pen.

3 6 friends share a bill equally and each pays \$23. Find the bill.

5 A taxi charges \$4 plus \$2 per km. A trip cost \$30. How far was it?

2 The perimeter of a square is 52 cm. Find the side length.

4 Tom is 3 times as old as Kate. Tom is 36. How old is Kate?

6 Movie tickets cost \$15 each plus a \$3 booking fee. The total was \$78. How many tickets?



Level 3 · Challenge

Finished Set D? Try the extension on the next page.

WE DO With your teacher: copy what they write on the board.**Example 5** The perimeter of a square is 52 cm.
Find the side length.
Example 6 Three consecutive numbers add to 72.
Find them.
YOU DO On your own: Set D. Harder problems.**1** Three consecutive numbers add to 72. Find them.
3 Raffle prizes cost \$180 and tickets are \$5. How
many tickets to make a \$2100 profit?
2 A rectangle is twice as long as it is wide. Its
perimeter is 48 m. Find its length and width.
4 Ali has \$50 more than Ben. Together they have
\$210. How much does Ben have?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show
your working.

Name _____ Class _____

Level 1A number plus 14 is 31. Write and
solve an equation.
Level 2Four equal books cost \$62. Find the
cost of one book.
Level 3A number is doubled and 9 is
added. The result is 41. Find the
number.

Extension · Stretch yourself

Worked example

A phone plan costs \$20 plus 10c per text. Another costs \$35 with free texts. How many texts make them cost the same?

1. $20 + 0.1t = 35$
2. $0.1t = 15$
3. $t = 150$

Answer: 150 texts

YOU DO

On your own: read the worked example, then try these.

- E1** A gym charges \$60 to join plus \$12 a week. Another charges \$20 a week with no joining fee. After how many weeks do they cost the same?

- E2** The sum of two numbers is 50 and their difference is 14. Find the numbers.

Lesson summary

Missed the lesson? Start here.

Remember 1. Choose a letter for the unknown.
3. Solve it step by step.

2. Write an equation from the words.
4. Answer the question in a sentence with units.

Level 1

Number $\times 17 = 1003$

1. $17N = 1003$
2. $\div 17$

Answer: $N = 59$

Level 2

Raffle: cover \$180

1. $5N = 180$

Answer: 36 tickets

Level 3

\$2100 profit

1. $5N - 180 = 2100$
2. $5N = 2280$

Answer: 456 tickets

Extension

Phone plans

1. $20 + 0.1t = 35$

Answer: 150 texts

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